Agricultural Crop Production

Comprehensive Data Analytics Project Report

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1. Introduction

Agriculture is one of the most important sectors of the Indian economy. Knowing which crops are produced the most, which are growing, and how production is linked with the area under cultivation and yield helps in planning and decision making. The objective of this project is to analyze the crop production data of India to understand production, area and yield of different crops from 2006-07 to 2010-11.

2. Dataset Description

Source: Kaggle, "Agriculture Crops Production in India"
Period: Five years, 2006-07 to 2010-11. Values are index numbers showing relative levels and changes, not actual quantities.

Column	Meaning
Crop	Name of the crop or crop group (e.g., Rice, Wheat).
Production	Production index of the crop in that year.
Area	Index of the area under cultivation.
Yield	Index of output per unit of area.
Category	Assigned crop group (Cereals, Pulses, Oilseeds, etc.).

3. Data Cleaning

- Checked missing values and duplicates: none found in any of the 16 columns.
- Cleaned crop names (e.g., correcting the missing space in "Rapeseed & Mustard").
- Validated Yield against (Production ÷ Area) × 100. Deviations in Tobacco and Sannhamp were noted but kept as given in the source.
- Group rows (e.g., "Total Foodgrains") were marked as Group Total so they are not double-counted against the 44 individual crops.

Chart 1: Crops by Category

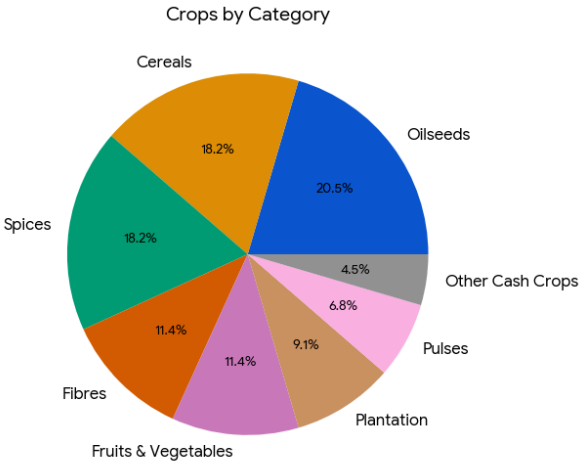


Figure 1: Oilseeds is the largest category with 9 of the 44 crops (20.5%), followed by Cereals and Spices (18.2% each). Other Cash Crops is the smallest (4.5%).

Chart 2: Top 10 Crops by Production (2010-11)

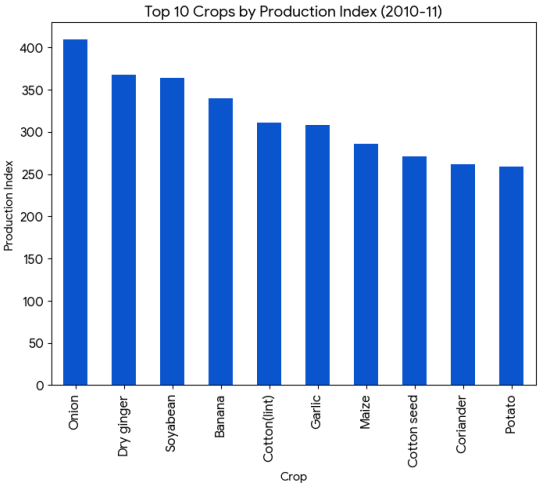


Figure 2: Onion has the highest production index at 409.1, followed by Dry ginger (368.1) and Soyabean (364.4). Spices, vegetables, and fibre crops dominate. Maize (286.0) is the only cereal in the top 10.

Chart 3: Average Production, Area and Yield by Year

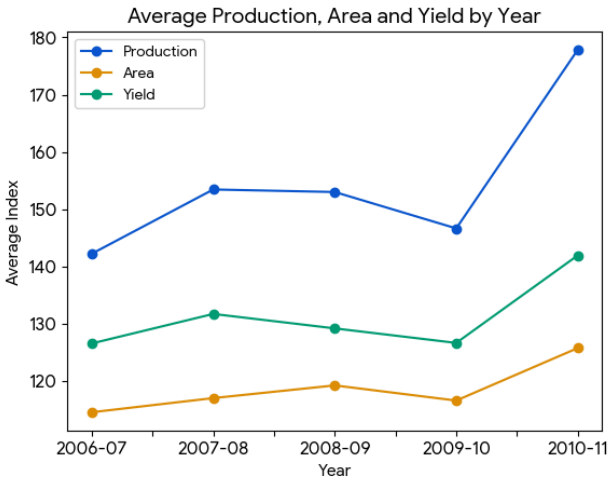


Figure 3: Average production index rose 25.0% from 2006-07 to 2010-11. Average area and yield moved in the same pattern, rising by 9.8% and 12.1% respectively, showing growth came from both more area and better yield.

Chart 4: Average Production by Category (2010-11)

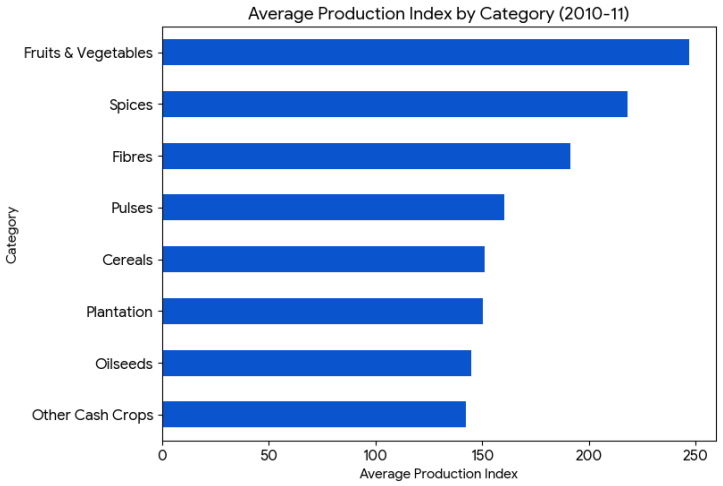


Figure 4: Fruits & Vegetables has the highest average production index at 247.2, followed by Spices (218.3) and Fibres (191.6). Other Cash Crops (142.6) and Oilseeds (145.1) have the lowest averages.

Chart 5: Top 10 Crops by Production Growth

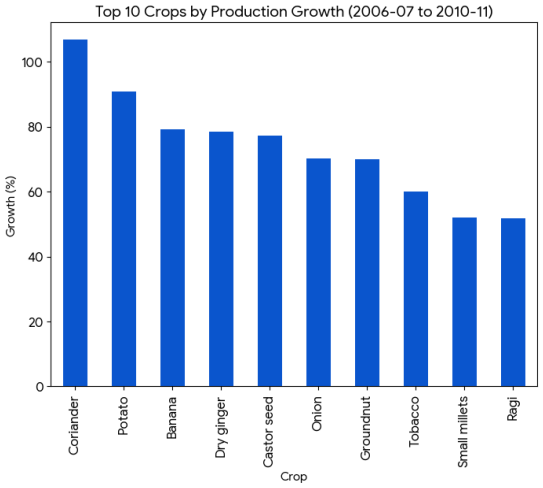


Figure 5: Coriander grew the most (+106.7%), followed by Potato (+90.9%) and Banana (+79.3%). Overall, 30 of the 44 crops grew and 14 declined. The biggest falls were in Sunflower (-46.9%) and Safflower (-37.4%).

Chart 6: Area vs Production (2010-11)

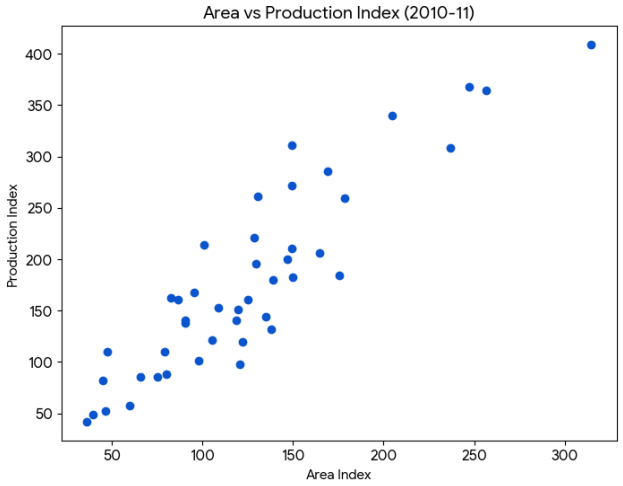


Figure 6: Crops with a larger area index tend to have a higher production index. The correlation is 0.90, which is a strong positive relationship. Onion stands out with both the largest area and highest production.

5. Key Findings

- The cleaned dataset has 55 rows: 44 individual crops and 11 group totals with no missing values or duplicates.
- **Oilseeds** has the most crops (9 crops, 20.5%), followed by Cereals and Spices (8 crops each).
- **Onion** has the highest production index in 2010-11 (409.1), followed by Dry ginger (368.1) and Soyabean (364.4).
- Between 2006-07 and 2010-11, the average production index rose by 25.0%, area index by 9.8% and yield index by 12.1%.
- 30 of the 44 crops increased production. Coriander grew the most (+106.7%) and Sunflower fell the most (-46.9%).
- Staple cereals grew slowly compared with other crops: Rice +2.8%, Wheat +14.6%, while Maize grew by 43.9%. Total Foodgrains grew 12.7%.

6. Recommendations

- **Support fast-growing crops:** Invest in better storage, transport, and market access for Onion, Potato, Banana, and spices so that higher production is not wasted.
- **Investigate declining crops:** Study why Sunflower, Safflower, Mesta, Coconut, and Black pepper declined (e.g., price, rainfall, or farmers shifting crops) and give targeted support to these groups.
- **Improve cereal yields:** Improve the yield of Rice and Wheat through better seeds and irrigation, as their production grew only marginally. Crops with low yield should be prioritized.

7. Conclusion

The analysis of 44 crops from 2006-07 to 2010-11 shows that crop production grew strongly over the period, with the average production index up 25.0%, helped by both a larger cultivated area and a higher yield. Fruits & Vegetables and Spices lead in overall production, while Cereals and Oilseeds are lower on average. Growth was uneven: most crops grew, but some, such as Sunflower and Coconut, declined sharply. Since area and production are strongly linked, increasing cultivated area works, but raising the yield of staple cereals and supporting fast-growing crops with infrastructure remains essential for future resilience and profitability.